In-domain

Methods	SQuAD		NewsQA		TriviaQA		SearchQA		HotpotQA		NQ		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
ComprExIT	50.46	66.86	23.10	39.03	55.35	61.59	64.08	70.72	50.50	68.97	44.38	61.97	47.98	61.52
Llama-3.2-3B														
ComprExIT	59.27	75.73	28.56	45.42	67.72	73.24	72.34	78.49	56.16	74.34	51.73	68.07	55.96	69.21
Llama-3.1-8B														
ComprExIT	65.43	80.04	29.80	47.77	74.25	79.39	78.69	84.35	59.87	77.11	54.91	71.38	60.49	73.34

Methods	SQuAD		NewsQA		TriviaQA		SearchQA		HotpotQA		NQ		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
Prompt tuning [w/ context]	71.89	81.09	30.72	45.76	61.04	66.75	64.03	70.64	53.60	69.49	53.03	66.59	55.72	66.72
Zero-shot [w/ context]	16.59	38.76	9.97	26.99	35.50	49.31	5.83	12.39	22.51	39.24	18.14	38.90	18.09	34.27
Zero-shot [w/o context]	2.80	8.83	0.97	3.11	15.63	21.60	4.04	6.64	2.76	7.17	3.37	8.43	4.93	9.29
ICAE [Ge et al., 2024]	36.84	50.21	20.20	33.73	57.59	63.54	67.40	74.43	40.67	57.04	42.26	58.00	44.16	56.16
500x [Li et al., 2025c]	4.65	14.02	1.85	5.27	26.09	32.66	17.44	25.59	5.56	13.44	5.58	13.89	10.20	17.48
Beacon [Zhang et al., 2025]	24.88	39.50	9.12	15.74	0.51	2.15	1.90	9.86	31.23	46.35	21.04	37.05	14.78	25.11
ComprExIT	**51.38**	**68.08**	**30.39**	**49.12**	**64.62**	**70.93**	**71.91**	**78.93**	**49.84**	**68.40**	**45.93**	**63.88**	**52.34**	**66.55**
Llama-3.2-3B														
Prompt tuning [w/ context]	80.39	88.41	35.23	50.98	71.23	76.30	72.04	78.42	55.31	72.65	59.46	73.32	62.28	73.35
Zero-shot [w/ context]	35.85	55.05	16.33	34.89	57.52	67.83	29.40	37.26	38.94	57.24	32.74	52.09	35.13	50.73
Zero-shot [w/o context]	9.14	17.21	2.40	5.78	46.20	52.06	25.43	30.35	8.63	16.03	10.31	18.47	17.02	23.32
ICAE [Ge et al., 2024]	48.51	62.53	27.02	43.04	68.62	74.13	76.43	82.89	47.92	66.16	50.87	66.75	53.23	65.92
Beacon [Zhang et al., 2025]	59.81	72.65	15.46	24.39	1.36	4.72	5.24	9.13	51.55	67.41	45.05	59.56	29.75	39.64
500x [Li et al., 2025c]	52.08	65.56	25.45	40.99	65.77	71.59	69.26	76.28	47.28	65.71	51.32	66.60	51.86	64.46
ComprExIT	**59.26**	**75.68**	**35.73**	**55.23**	**72.97**	**78.86**	**78.00**	**84.40**	**55.92**	**74.15**	**52.23**	**68.95**	**59.02**	**72.88**

Out-of-domain (OOD)

Methods	RelationExtraction		BioASQ		TextbookQA		DuoRC		DROP		RACE		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
ComprExIT	61.02	75.55	47.81	63.46	41.85	50.70	28.25	37.20	30.67	41.81	22.55	36.34	38.69	50.84
Llama-3.1-8B														
ComprExIT	67.88	80.49	63.90	78.98	59.95	69.13	39.51	49.93	46.97	56.91	35.46	50.59	52.28	64.34

Methods	RelationExtraction		BioASQ		TextbookQA		DuoRC		DROP		RACE		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B-Base														
Prompt tuning [w/ context]	69.91	81.15	65.96	78.80	52.10	59.61	38.51	47.40	32.54	43.14	33.68	46.30	48.78	59.40
Zero-shot [w/ context]	27.48	47.87	6.98	28.84	18.16	29.97	17.46	30.54	18.43	32.30	10.53	22.71	16.51	32.71
Zero-shot [w/o context]	5.09	10.33	14.76	20.56	11.11	16.03	0.60	2.55	14.84	19.73	0.45	2.88	7.81	12.01
ICAE [Ge et al., 2024]	**62.28**	**75.97**	42.62	54.09	28.21	34.60	12.59	18.80	24.15	33.23	9.50	16.95	29.89	38.94
500x [Li et al., 2025c]	6.31	16.30	22.54	35.52	16.77	24.40	1.07	3.91	8.18	18.32	1.93	5.70	9.47	17.36
Beacon [Zhang et al., 2025]	33.18	48.82	32.25	47.84	5.92	10.34	2.40	4.90	21.82	29.24	15.88	25.35	18.57	27.75
ComprExIT	59.46	74.96	**48.40**	**63.74**	**41.65**	**50.73**	**28.78**	**38.77**	**29.01**	**39.86**	**23.00**	**37.95**	**38.38**	**51.00**
Llama-3.2-3B-Base														
Prompt tuning [w/ context]	77.37	86.70	67.62	81.50	60.48	68.75	45.50	55.63	48.84	57.18	43.62	56.35	57.24	67.68
Zero-shot [w/ context]	54.24	68.46	43.22	64.31	36.46	46.66	28.31	42.36	23.69	37.45	22.70	38.16	34.77	49.57
Zero-shot [w/o context]	14.28	21.03	20.94	25.76	23.29	28.89	2.07	4.44	16.17	21.26	1.78	5.65	13.09	17.84
ICAE [Ge et al., 2024]	**70.66**	**82.10**	55.85	67.21	42.25	50.70	23.65	32.81	35.33	44.89	21.51	33.85	41.54	51.93
500x [Li et al., 2025c]	65.13	79.67	47.67	60.82	29.41	38.10	22.25	30.91	29.74	39.12	19.14	30.58	35.56	46.53
Beacon [Zhang et al., 2025]	58.92	71.62	54.52	66.86	10.98	15.68	4.73	7.95	39.39	50.55	30.71	43.95	33.21	42.77
ComprExIT	65.31	77.25	**59.64**	**73.87**	**56.09**	**64.50**	**34.31**	**44.98**	**43.58**	**53.16**	**31.60**	**47.69**	**47.19**	**59.34**

Llmma3.2 1B comprexit

数据集	模式	N	EM %	token F1 %	平均秒/题	截断上下文数
countries	compressed	200	39.50	40.58	0.153	0
countries	uncompressed	200	15.00	19.52	0.674	0
tipsheets	compressed	500	23.20	53.37	0.175	0
tipsheets	uncompressed	500	12.20	23.69	0.192	0

In-domain

Methods	SQuAD		NewsQA		TriviaQA		SearchQA		HotpotQA		NQ		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
ComprExIT	50.46	66.86	23.10	39.03	55.35	61.59	64.08	70.72	50.50	68.97	44.38	61.97	47.98	61.52
Llama-3.1-8B														
ComprExIT	65.43	80.04	29.80	47.77	74.25	79.39	78.69	84.35	59.87	77.11	54.91	71.38	60.49	73.34

Out-of-domain (OOD)

Methods	RelationExtraction		BioASQ		TextbookQA		DuoRC		DROP		RACE		Average	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Llama-3.2-1B														
ComprExIT	61.02	75.55	47.81	63.46	41.85	50.70	28.25	37.20	30.67	41.81	22.55	36.34	38.69	50.84
Llama-3.1-8B														
ComprExIT	67.88	80.49	63.90	78.98	59.95	69.13	39.51	49.93	46.97	56.91	35.46	50.59	52.28	64.34